

Background

Compact high-redshift sources overlap with any host light inside one diffraction-limited image. Astronomers subtract a point-spread function built from nearby field stars to isolate the extended component.

That stellar reference assumes the target's spectrum matches the stars, which little red dots violate.

What this paper is selling

Using little red dots as a test case, the authors show broad-filter stellar PSFs systematically misassign light, inflating or inventing host flux. They derive the physical origin of the error and demonstrate a source-conditioned polychromatic correction.

Key results to remember

- A pure point source yields spurious Sérsic residuals in every filter.
- The published 11.8% F444W host component is exceeded by all tested configurations.
- Inferred sizes, luminosities and masses of compact nuclei are biased high.

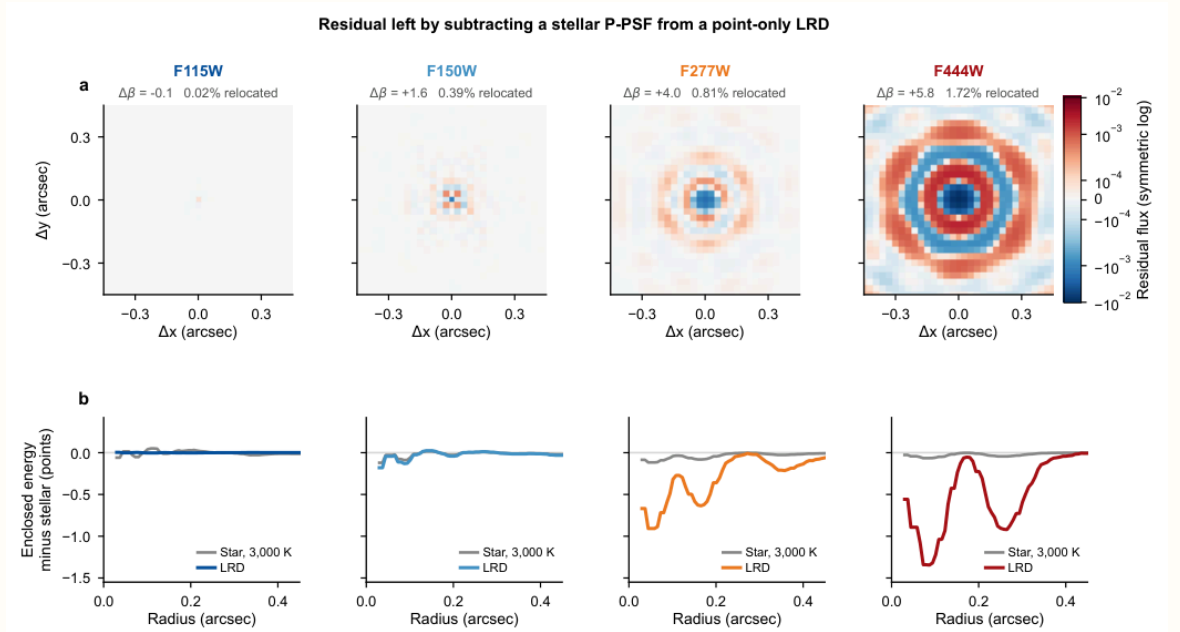


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves wavelength-dependent residuals that mimic a Sérsic host.

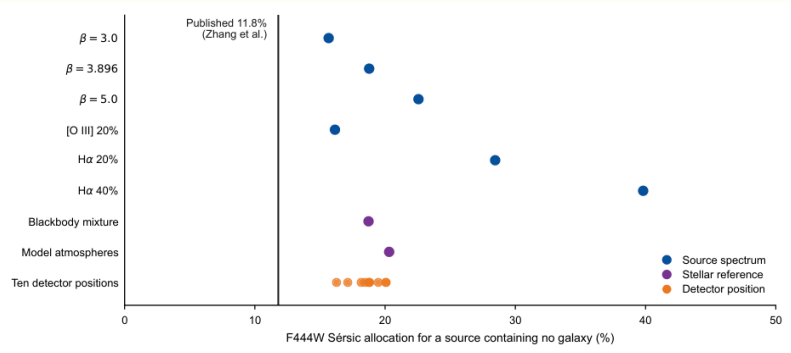


Figure 2 | Every tested nuclear spectrum, stellar reference and detector position allocates more F444W flux than the published 11.8% host.